THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET A

Chapter 2: **Exploring Mixtures and Their Separation** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. 10 g of salt is dissolved in 90 g of water. The mass by mass percentage of the solution is: 1

(a) 11.11 % **(b)** 20 % **(c)** 5 % **(d)** 10 %

2. 5 g of a solute is dissolved in water to make 100 mL of solution. Its mass by volume percentage (% m/v) is: 1

(a) 5 % **(b)** 10 % **(c)** 4.76 % **(d)** 2.5 %

3. A homogeneous mixture of two or more substances with uniform composition throughout is called a: 1

(a) heterogeneous mixture **(b)** precipitate
(c) solution **(d)** suspension

4. A solution that cannot dissolve any more solute at a given temperature is called: 1

(a) a colloid **(b)** a suspension
(c) a saturated solution **(d)** a dilute solution

5. The separation of crude petroleum into petrol, kerosene, diesel etc. uses: 1

(a) sublimation **(b)** fractional distillation
(c) chromatography **(d)** crystallization

6. Which of the following sublimes on heating? 1

(a) iron filings **(b)** sand **(c)** common salt **(d)** naphthalene

7. The scattering of a beam of light by particles in a colloid is called the: 1

(a) Tyndall effect **(b)** Raman effect
(c) Doppler effect **(d)** photoelectric effect

8. A homogeneous mixture of two or more metals, or a metal and a non-metal, is called: 1

(a) a colloid **(b)** a suspension **(c)** a compound **(d)** an alloy

9. The concentration of normal saline used in hospitals is about: 1

(a) 0.09 % m/v **(b)** 90 % m/v **(c)** 0.9 % m/v **(d)** 9 % m/v

10. ‘Mitti ka ittar’ of Kannauj is prepared using the traditional technique of: 1

(a) sublimation **(b)** distillation **(c)** crystallization **(d)** chromatography

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Define concentration of a solution. Name two ways of expressing it. 2

12. Calculate the mass by mass percentage if 10 g of salt is dissolved in 90 g of water. 2

13. 5 g of glucose is dissolved in water to make 100 mL of solution. Find its % m/v. 2

14. Why is the temperature always mentioned while stating the solubility of a substance? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Calculate how much zinc oxide is present in 300 g of a talcum powder containing 4 % m/m zinc oxide. 3

16. A glass is filled with 150 mL of orange drink prepared using two tablespoons of concentrate (15 mL each). Find the % v/v of the concentrate. 3

17. Vinegar contains 5 % v/v acetic acid. How much glacial acetic acid is needed to make 200 mL of vinegar? 3

OR

A saline solution is 0.9 % m/v. How much salt is present in 500 mL of the solution? 3

18. What do solubility curves show? State two inferences from a curve in which solubility of compound B increases sharply with temperature. 3

OR

Why should a hot saturated solution of copper sulfate be filtered before cooling? Why is a drop of dilute sulfuric acid added? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Define % m/m, % m/v and % v/v.
(ii) 25 g of sugar is dissolved in 225 g of water; find % m/m.
(iii) How much solute is there in 250 mL of a 4 % m/v solution? 5

OR

Case study: A farmer must prepare a spray from a liquid pesticide for rice. 5
(i) Why must the proportion be right?
(ii) If 1 mL is mixed to make 100 mL, find % v/v.
(iii) How much pesticide is needed for 20 L of spray at this strength?
(iv) Name the type of mixture.

20. (i) What is crystallization? 5
(ii) Describe how pure crystals of copper sulfate are prepared in the laboratory, stating the precautions.
(iii) How do crystals form in nature (give one example)?

OR

Case study: The solubility of two compounds A and B is plotted against temperature. A shows a small rise and B a large rise as the temperature increases from 20 °C to 80 °C. 5
(i) Which dissolves more at 60 °C?
(ii) If saturated solutions of equal mass of A and B at 80 °C are cooled to 40 °C, which deposits more crystals? Why?
(iii) Which compound can be purified by crystallization more efficiently?

END OF QUESTION PAPER — **Total: 40 Marks**